

BUIDL CTC 2026 FALL · RWA

Proofmark

Verify once. Carry the result to every chain.

A KYC and AML attestation layer. Screening results are issued on Ethereum, verified on Creditcoin through the Attestcoin Protocol with no oracle operator in between, and read by any chain that wants them.

ON-CHAIN PERSONAL DATA

0 bytes

SANCTIONS ENTRIES SCREENED

26,566

OFAC, UN and EU, from source

ISSUANCE TO VERIFIED

7m 55s

measured, Sepolia to Creditcoin

THE PROBLEM

Compliance splits twice. Once per chain, once per border.

Every service inside a regulatory perimeter opens with the same question: **may I transact with this wallet?**

There are three ways to answer it today.

OPTION 1

Rebuild KYC per chain

Vendor minimums, months of integration and data-controller obligations multiply by the number of chains. The user uploads the same document again.

OPTION 2

Trust a signing server

One key becomes the whole security model, and there is no good answer to a regulator asking why that signature should be believed.

OPTION 3

Trust a relay

Trusting the relay becomes trusting the compliance. The trust moved rather than disappeared.

Crossing a border adds a second problem. **"KYC complete" names different work in different places.**

JURISDICTION	WHAT IT ACTUALLY REQUIRES
Korea	At least two of: document image, video call, verification on delivery of an access device, use of an existing account. Plus CDD under the FIU act.
EU	AMLD customer due diligence, graded against an eIDAS assurance level.
US	CIP: collect and verify name, date of birth, address and an identification number.

A single boolean satisfies one of them and gives the others nothing to stand on.

The mark carries the checks, not the verdict.

Sixteen bits record what actually happened. Each consumer applies its own policy to those bits and reaches its own conclusion. We publish evidence and assert no equivalence.

IDENTITY CHECKS

Wallet control	Document image
Document authenticity	Face match
Liveness	Bank account
Mobile carrier	ePassport NFC

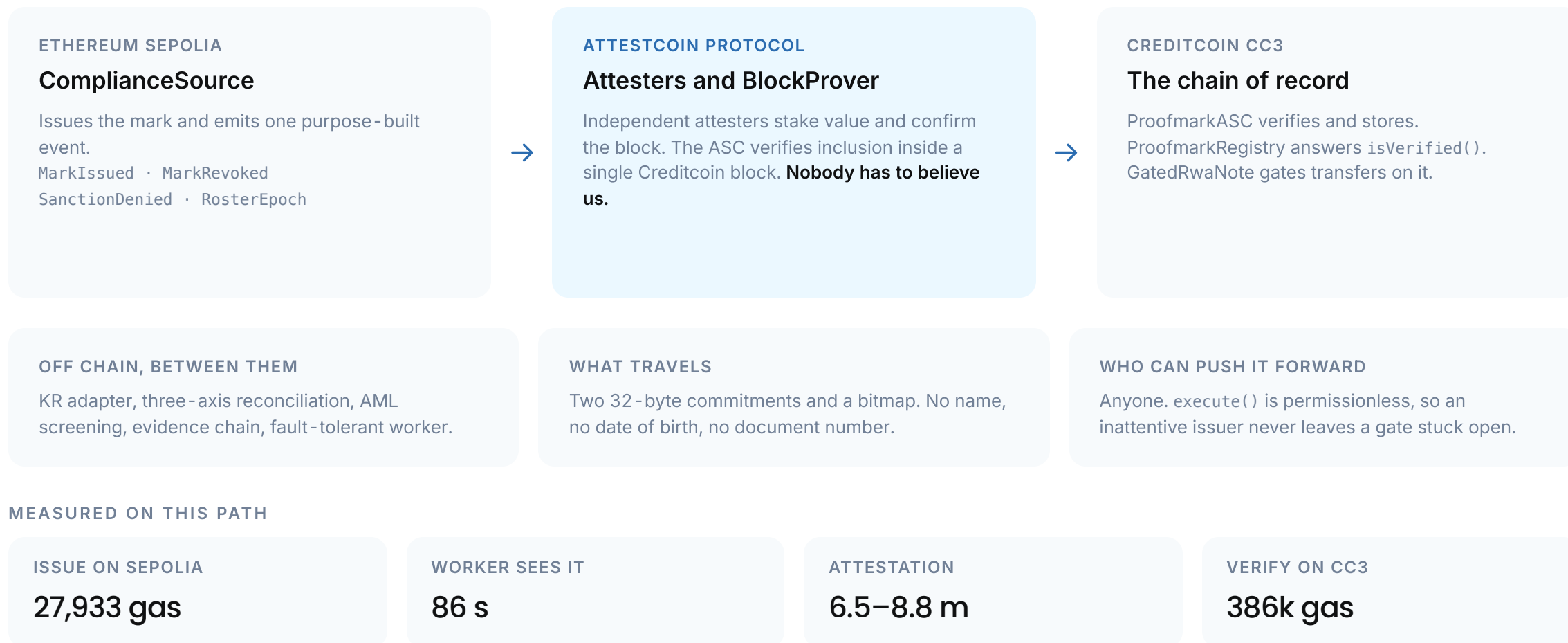
SCREENING

Sanctions	PEP
Adverse media	Jurisdiction
On-chain exposure	

A Korean VASP requires document authenticity, bank account and sanctions. An EU issuer requires liveness, sanctions and PEP. **The same mark answers both questions differently, and neither answer is ours.**

ALSO ON THE MARK	WHY
regime	Which framework the checks ran under
jurisdiction	Where it was issued
assurance	The issuer's own grade. Not a claim of equivalence
claimsRoot	Salted commitment. The values stay in the browser
evidenceHash	Head of the screening hash chain

Issued on Ethereum. Proven on Creditcoin. Read anywhere.



Remove Attestcoin and see what survives.

REMOVE IT	WHAT IS LEFT
Creditcoin cannot learn what Ethereum issued	A relay we operate, which is the trust the product exists to remove
Epoch roster roots cannot be checked	Issuer signatures, back to trusting one key
A spoke chain's mirror cannot be disproven	Trusting the mirror operator

Attestcoin is not a feature of this product. It is the trust model.

Two security holes that inheritance cannot close.

`ASCBASE.execute()` receives `chainKey` and `blockHeight` and passes neither to the handler. We forked it. Verification, `queryId` and replay protection are untouched.

FINDING 1 · CHAINKEY SPOOFING

CC3 serves Sepolia (`chainKey` 1) and Ethereum mainnet (3) at the same time. A handler that only checks `log.address_` accepts a forged event from a same-address contract on the other chain, and `CREATE2` makes claiming that address first straightforward.

```
require(chainKey == expectedChainKey)
```

FINDING 2 · REORDERING REVIVES A DEAD MARK

Submission is permissionless and unordered. Send `MarkIssued` from block 100 after `MarkRevoked` from block 200 and the mark returns to `ACTIVE`. The `queryIds` differ, so replay protection never fires. Both proofs are valid.

```
lastAppliedHeight[subject]
```

Both guards are **mutation tested**. Remove either one and exactly its test fails. `test_RejectsProofFromWrongChain` · `test_StaleIssueCannotResurrectRevokedMark`

We also found `verifyBatch` has no contract-side entry point, so batching runs through multiple logs in one transaction instead. One `execute()` applies all of them.

MEASURED, NOT ESTIMATED

What the round trip actually costs.

ISSUANCE TO VERIFIED

7m 55s

and 10m 48s on the second run

ATTESTATION LAG

6.5–8.8m

three observations

PROOF VERIFICATION

386k

gas on CC3

COST PER QUERY

0.0002

CTC, gas only

We do not describe revocation as instant. Source chain finality sets a floor, so the number is published as a product parameter. Anything needing real-time blocking gates on the source chain.

The measurement also corrected us twice. The faucet gave 10,000 CTC rather than the documented 100, and a query costs gas alone with no protocol fee. Both facts removed arguments we had already written down.

STEP	VALUE
Sepolia issuance	27,933 gas
Worker sees the event	86 s
Attestation completes	6m 30s
CC3 proof verification	386,008 gas
isVerified turns true	7m 55s
Revocation propagates back	8m 43s

SCREENING

Real lists. 26,566 entries. No fixtures.

RECALL

100%

200 listed individuals, looked up by their own details

SPECIFICITY

100%

610 ordinary names, zero false positives

EVASION

7 / 7

homoglyphs, invisible characters, diacritics, word order

SANCTIONED WALLET

124

EVM addresses from the OFAC list

Romanised expansion is our inference, not something a list asserts. That distinction decides the product.

INPUT	DECISION	WHY
Kim Jong Un, KP, DOB matches	BLOCK band 5	Expansion hit, corroborated by date of birth and country
Choi Yeong-ho, KR	ALLOW band 2	Expansion hit against the listed Choi Yong-ho, and nothing corroborates it

Strip the diacritics and both names become yong, scoring 100 against each other. Letting expansion drive decisions produced a **33% false positive rate** on ordinary Korean names. Refusing to let it produces zero, and Kim Jong Un is still caught.

THE MECHANISM

The mark we issue does not pass the production policy we wrote.

That policy asks for document authenticity and bank account verification. Those need institutional agreements we do not have, so those checks did not run, so those bits are not set, and the system reports it.

KR VASP production

policyId 1 · requireAll 0x10024

Missing: Document authenticity, Bank account

FAIL

KR pilot

policyId 2 · requireAll 0x190001

Wallet control · Sanctions · Jurisdiction · On-chain exposure

PASS

We did not lower the production policy to fit our own mark. An unconnected check is a zero bit, and a zero bit is what lets a consumer reject the mark. Both cards are read live from Creditcoin.

Reproducible in five minutes.

Every address and transaction below is live. The README carries the commands.

CONTRACT	CHAIN	ADDRESS
EvmV1Decoder	CC3 Testnet	0xff3558704c75ed69e1D657474210365b24d31938
ProofmarkASC	CC3 Testnet	0x93C62D3016123Da0aBdB4AC1857564c30CbE5629
ProofmarkRegistry	CC3 Testnet	0x874e0Fd030a8Fe6c7a06835354531b68A31f5FCc
ComplianceSource	Ethereum Sepolia	0x93C62D3016123Da0aBdB4AC1857564c30CbE5629
GatedRwaNote	CC3 Testnet	0xA8Dfe6f5063a8159524DeddBAC75f034C3BF0625

TESTS

45 + 93

Solidity and TypeScript, all passing

ISSUANCE TRANSACTION

0xa251db9d

Sepolia block 11,602,963

PROOF SUBMISSION

0x8e96ce1f

CC3, 388,696 gas

One thing the README states twice: pass --from to cast call. Without it msg.sender is zero, onlyOwner fires before the gate, and you read the wrong revert. We walked into it ourselves.

Who buys this, and what they do without it.

CUSTOMER	WHY THEY BUY	WHAT THEY DO INSTEAD	PRICING
RWA and stablecoin issuers	A whitelist runs every day and produces evidence for a supervisor	Stand up an operations team and build the evidence system themselves	Annual contract with an SLA
Cross-chain dApps	One integration instead of one per chain	Vendor contracts multiplied by chain count	Per issuance. Queries free
Multinational dApps	Change the policy, not the plumbing, when a jurisdiction changes	Jurisdictions times vendors, in integrations	Adapter subscription
Wallets	Check a recipient before sending	Incident response, and explaining that funds cannot be recovered	Per query
VASPs and custodians	Pre-check personal wallet recipients, travel rule	Manual review and held withdrawals	Per check

What this gives Creditcoin. Issuers legally required to screen holders gain a reason to read it, and Attestcoin is the only trustless cross-chain read there is.

What we will not claim. Attestcoin reads are free, and this product sits entirely on that path. Saying we generate ATC demand would be false, so we do not say it.

People who ran this compliance stack before, off chain.

Seven years operating Korea's fourth registered VASP, GDAC, covering the CEO, compliance officer and CTO roles, plus the engineer who built an exchange's KYC pipeline.

NAME	ROLE	BACKGROUND
Ash Han	CEO	Founded and ran GDAC for 7 years. Secured Korea's 4th VASP registration, ISMS certification and FinCEN MSB. Advisory director at Tendermint, founding member of the Interchain Foundation.
Youree Lee, Ph.D	Chief Compliance Officer	CCO and deputy CEO at Peertec, GDAC's operator. Set AML and KYC policy for a licensed exchange and handled FSC examinations. Previously Shinhan Macquarie and Sumitomo Mitsui Banking.
Yongku Lee	CTO	CTO at Peertec, at Riiid and at Permachain. Exchange core, wallet and custody, security systems, at production scale.
Incheol Yang	Product and protocol lead	Engineer at Coinone on exchange core and non-face-to-face KYC integration. Co-founded Baekdu Technologies, leading Solidity work on stableswap, leverage and perp DEX protocols. Wrote the code deployed here.

Top rating in the FSC's money laundering risk assessment for three consecutive quarters, ranking in the top five of 35 registered VASPs. [That record belongs to GDAC, where the work was done. Validator Inc. started in 2026 and does not hold those licences.](#)

The next jurisdiction is a passport chip.

NEXT	WHEN
Epoch rosters published on chain, which is how "not revoked" gets proven	next
ePassport NFC adapter. The chip is signed by the issuing country, so it verifies with no vendor and no jurisdiction dependency	after
Vendor integration for document authenticity and bank account	needs agreements
Spoke chain mirrors, then push once Attestcoin writability ships	roadmap

WHAT DOES NOT WORK YET	
Document authenticity, face match, liveness, bank account	no vendor
PEP and adverse media screening	no data licence
FATF jurisdiction table checked against the source	unverified
Writability, so no push to spoke chains	protocol

Each of these is a zero bit on the mark, which is why the list can be printed on a slide without costing anything. `pipeline/integration.test.ts` also proves the reverse: connect the vendors and the production policy passes.

LIVE DEMO
attest-kyc.stabled.ai

REPOSITORY
github.com/stabled-ai/
attest-kyc-ctc

LIVE ON
CC3 Testnet · Ethereum Sepolia

CONTACT
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